SOIL EROSION AND CONSERVATION · NOTE ST07

Drop, Chute and Drop-Inlet Spillways — Design of Inlets, Box-Inlet Outlets, SAF Basins and Pipe Spillways

Course topics: SWCE-503 T8, SWCE4.5 L13, SWCE8.12 L8, SWCE-501 T12

The three principal gully-control spillways

Drop, chute and drop-inlet spillways are grouped by inlet, conduit and outlet. Most have four components: an **earthen embankment** (checks flow and directs it into the spillway, extending to high ground), an **inlet** (weir, box — water entering from three or four sides, straight, flared or curved walls — or culvert entrance: round, square or flared), a **conduit**, and an **outlet** that releases water at non-erosive velocity (cantilever outlet where the downstream channel is unstable, plain apron, or apron with energy dissipators).

- **Site:** chosen from field survey, hydrology, structural and hydraulic needs and economics — on a straight channel reach of 25–50 m with no curve upstream or downstream, and a foundation with adequate bearing capacity, resistance to sliding and piping, and homogeneity.
- **Design runoff:** permanent structures for a 50-year recurrence interval, after studying watershed, rainfall, soil and geology, and the downstream channel.

Drop spillway

- Water flows through a weir opening, falls onto a level apron or stilling basin, dissipates its energy and enters the downstream channel. It stabilises the channel profile by turning a steep gradient into gentle reaches, and serves as an outlet for tile and surface drainage.
- **Maximum drop 3–4 m** — larger drops need a chute or drop-inlet spillway (a 4 m drop is not economical). Built of stone or brick masonry, concrete or concrete blocks: very stable, little structural damage, easy, economical for low and medium drops, easily standardised; a rectangular weir inlet clogs less. Not suitable where temporary storage is needed.
- **Failure:** insufficient hydraulic capacity or structural failure.
- **Inlets:** straight (wide, shallow gullies, small to medium flows), box (large flows, narrow structure, low head), curved/arched (like a box, with arch strength).

Weir formula: $Q = (2/3)\cdot C_d\cdot\sqrt{(2g)}\cdot L\cdot H^{(3/2)}$ Metric: $Q = 1.77\cdot L\cdot H^{(3/2)}$ (m^3/s, m)
L = crest length (three sides of a box inlet, circumference of an arch inlet, or straight crest); freeboard 15 cm above the computed head

- When tailwater reaches the crest, capacity falls with the degree of submergence.
- **Outlet:** the falling water's energy must be dissipated or converted. A stilling basin is preferred to a straight apron — shorter, with the jump at the toe; longitudinal sills straighten flow and suppress transverse velocity.

Box-inlet drop spillway outlet (Blaisdell and Donnelly, 1951)

A rectangular box open at the top and downstream end; water enters at the upstream end and sides and leaves downstream, guided by dikes and headwalls.

$$\text{Critical depth, straight section: } d_c = \sqrt[3]{(Q/W)^2/g} \text{ at exit: } d_{ce} = \sqrt[3]{(Q/W_e)^2/g}$$
$$\text{Straight-section length: } L_s = d_c \cdot (0.2/(B/W) + 1) \text{ for } B/W > 0.25 \text{ (use larger lengths)}$$
$$\text{Stilling-basin length (as printed): } L_B = L/(2B/W)$$
$$\text{Tailwater, basin width} > 11.5\, d_{ce}\text{: } d_3 = 1.6\, d_{ce} \text{ (from the end sill: } 1.33\, d_{ce})$$
$$\text{Tailwater, otherwise: } d_3 = d_{ce} + 0.052\, W_e \text{ (from the end sill: } 0.83\, d_{ce} + 0.043\, W_e)$$
$$\text{End-sill height } f = d_2/6 \text{ (longitudinal sills the same); side-wall height above water } t \geq d_2/3$$

- **Wingwalls:** triangular in elevation with a 45° top slope (30° permissible); flared 60° in plan (45° permissible). Wingwalls parallel to the centre line are not recommended.

Structural design

- Consider sliding, piping, uplift, undermining, fill slopes and lateral scour. The **headwall** holds a stable fill, prevents piping, reduces uplift and resists sliding; the **toewall** prevents piping and undermining; **sidewalls** hold the fill and prevent erosion; **wingwalls** hold the fill and prevent scour.
- Common structural failures: overturning, sliding, tension and crushing. Stability is checked by stability factors (sliding, shear friction, overturning) and stress factors (no tension).

$$\text{Sliding factor: } SF = \Sigma H/(\Sigma W - U) \leq 0.8 \text{ for concrete, } 0.75 \text{ for masonry}$$
$$\text{Shear-friction factor: } Q = [(\Sigma W - U)\cdot f + A\cdot S]/H \geq 5$$
$$\Sigma H = \text{horizontal forces; } \Sigma W = \text{vertical forces; } U = \text{uplift; } f = \text{coefficient of friction; } A = \text{area of section; } S = \text{shear resistance}$$

Overturning factor (stabilising ÷ overturning moment): 1.2 for concrete and 2.5 for masonry dams (as printed). Choose the base width so the resultant passes through the downstream middle-third point with the reservoir full — no tension.

Chute spillway

- A lined open channel on a steep slope with built-in inlet and outlet; flow is supercritical. Components: inlet, vertical-curve section, steep channel, outlet.
- Used with earth dams and wherever water must drop farther than a drop structure can economically manage, a large discharge is needed, or temporary storage is impossible. It needs less masonry or concrete than a drop structure, has high capacity, and outlet sedimentation does not reduce its capacity. Risks: rodents undermining it and seepage endangering the foundation in poorly drained sites. Build on solid ground or carefully compacted fill in reinforced concrete or brick or stone masonry.
- **Inlet** (straight, box or side-channel; designed like a drop-spillway inlet) controls capacity, conveys and guides the design flow and gives a cutoff against piping under and around the chute.
- **Channel:** the vertical curve guides flow in; the channel is usually rectangular with straight, parallel side walls (tops flush with the embankment slope) for good velocity and discharge distribution, designed by Manning's formula to carry the flow down without erosion.
- **Outlet:** dissipates energy by a hydraulic jump and leaves a non-erosive velocity, on drop-spillway principles.

$$\text{Sequent depth ratio: } y_2/y_1 = \tfrac{1}{2}\cdot(\sqrt{(1 + 8F_n{}^2)} - 1)$$

SAF stilling basin (F_n = 1.7 to 17)

Item	Criterion
Basin length	L_B = 4.5·y_2/F_n^0.76
Chute blocks	Height y_1; width and spacing 0.75·y_1
Floor blocks	L_B/3 from the upstream end, downstream of the chute-block openings, occupying 40%–55% of basin width (widths and spacings grow in proportion in a diverging basin)
Side-wall clearance	At least 3y_1/8
End sill height	c = 0.07·y_2
Tailwater depth y_2'	F_n 1.7–5.5: (1.10 - F_n^2/120)·y_2; F_n 5.5–11: 0.85·y_2; F_n 11–17: (1.00 - F_n^2/800)·y_2
Side-wall height	z = y_2/3 above maximum tailwater
Wingwalls	Same height as basin sidewalls, top slope 1 : 1, at 45° to the outlet centre line; basin sidewalls parallel (rectangular) or diverging (trapezoidal); a nominal cutoff wall at the end; air entrainment neglected

*Corrections (checked against the Saint Anthony Falls criteria): the basin-length exponent is **0.76**, not 0.176; and the first tailwater case uses **F_n^2**, not F_n — with the square, the three cases agree at their boundaries (1.10 - 5.5²/120 = 0.85 and 1.00 - 11²/800 = 0.85).*

Drop-inlet spillway

- Ideal when downstream flow must be controlled and temporary storage is wanted upstream. It is an earth dam (temporary storage) plus a pipe spillway (passes the design discharge): a vertical **riser** on the upstream side connected to a pipe or culvert through the dam, then an outlet. The crest level is fixed to stabilise the upstream grade and protect the gully head; the inlet or the conduit controls flow.
- **Uses:** headwater flood control, farm ponds and reservoirs, silt-detention reservoirs, settling basins; gully control where the gully is deeper than 3 m. Relatively simple to build; temporary storage reduces the required capacity, the cost and the downstream peak, and helps stabilise grades.
- **Design:** the inlet by the weir formula; the pipe by the Darcy equation. At first the inlet (weir flow) controls discharge, later the pipe capacity does.

$$\text{Pipe velocity (Darcy, friction only): } V = \sqrt{(2g\cdot H\cdot d/(f\cdot l))}$$

$$\text{Discharge: } Q = a\cdot v$$

$$V \text{ in cm/s; } H = \text{available head (cm); } d = \text{pipe diameter (cm); } g \text{ in cm/s}^2\text{; } l = \text{pipe length (cm); } f = \text{Darcy friction factor; } a = \text{pipe area (cm}^2\text{); } Q \text{ in cm}^3\text{/s } (\div 1000 \text{ for litres/s})$$

Correction: the notes print V = √(Hdg/(2fl)). From the Darcy loss H = f·(l/d)·V²/2g the factor 2 belongs in the numerator: V = √(2gHd/(fl)). Entrance and exit losses are ignored here.

- **Emergency spillway:** round one end of the dam to pass rare high floods — good vegetation, uniform section, gradual downstream slope, no abrupt turns; its level lets the drop inlet reach its rated

capacity before it operates.

- **Outlet:** at or slightly above the downstream channel grade. A **propped outlet** suits drop-inlet spillways: the pipe extends 2 to 2.5 m beyond the pipe end, clear of the scour hole, supported on a masonry or concrete pillar, about 2.5 m downstream of the dam toe, so water lands safely away from the structure. Do not use it where the downstream grade is steeper than the limit given ("> 1" in the notes) without a supporting structure; alternatives are a straight-apron or stilling-basin outlet.

Illustration (not in the notes)
Pipe d = 30 cm, l = 2000 cm, H = 300 cm, f = 0.02, g = 981 cm/s²
V = √(2 × 981 × 300 × 30/(0.02 × 2000)) = √(17,658,000/40) ≈ 664 cm/s
Q = (π × 30²/4) × 664 ≈ 707 × 664 ≈ 469,000 cm³/s ≈ 469 L/s

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. US Bureau of Reclamation, Design of Small Dams and Hydraulic Design of Stilling Basins and Energy Dissipators (EM 25). Blaisdell, F. W. (1949), The SAF stilling basin, USDA-SCS SCS-TP-79. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Maximum drop of a drop spillway is about
(a) 1 m (b) 3–4 m
(c) 10 m (d) 20 m

Q2. Drop spillways should be on a straight reach of
(a) 5–10 m (b) 25–50 m
(c) 100 m (d) 500 m

Q3. Permanent gully-control spillways are designed for a recurrence interval of
(a) 5 yr (b) 10 yr
(c) 25 yr (d) 50 yr

Q4. SAF stilling basins perform for Froude numbers of
(a) 1–1.7 (b) 1.7–17
(c) 17–30 (d) > 30

Q5. SAF basin length is
(a) $4.5y_2/F^{0.76}$ (b) $4.5y_2F$
(c) $6y_2$ (d) $10y_1$

Q6. SAF end-sill height is
(a) $0.07y_2$ (b) y_1
(c) $y_2/3$ (d) $0.5y_2$

Q7. A drop-inlet spillway is suited to gullies deeper than
(a) 0.5 m (b) 1 m
(c) 3 m (d) 10 m

Q8. Sliding factor limit for concrete structures is
(a) 0.5 (b) 0.75
(c) 0.8 (d) 1.5

Q9. Minimum shear-friction factor is
(a) 1 (b) 2
(c) 5 (d) 10

Q10. Darcy pipe velocity is

(a) √(2gHd/(fl)) (b) √(Hdg/(2fl))

(c) 2gH/fl (d) √(2gH)

Answers

Q1 — (b) 3–4 m

Q2 — (b) 25–50 m

Q3 — (d) 50 yr

Q4 — (b) 1.7–17

Q5 — (a) $4.5y_2/F^{0.76}$

Q6 — (a) $0.07y_2$

Q7 — (c) 3 m

Q8 — (c) 0.8

Q9 — (c) 5

Q10 — (a) √(2gHd/(fl))
